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Chloride Ion Effects on Passive Films on FeCr and FeCrMo Studied by AES, XPS and SIMS

D. Landolt, S.Mischler, A.Vogel, H.J.Mathieu
Materials Department
Swiss Federal Institute of Technolgy
Lausanne /Switzerland

Introduction

Surface analysis methods such as AES, XPS and to a lesser extent SIMS have been applied to the study of passive oxide films for a number of years and valuable information on chemical composition and film thickness has been obtained. A recent review of the literature (1) shows, however, that published surface analysis data of passsive films do not always agree, in particular those concerning the presence or not of chloride ions in the films. *Fig 1* gives a summary of published results (1).
A better understanding of the composition of passive films on iron-chromium alloys is of particular interest because of the practical importance of stainless steels as corrosion resistant alloys. Molybdenum is a common alloying element in stainless steels. Its presence shifts the pitting potential to more noble values and thus increases the resistance of the alloy to pitting corrosion. The reaction mechanism of molybdenum is still not well known, however.

In our laboratory the surface analysis methods AES, XPS and SIMS are being used for the investigation of passive films on iron-chromium and iron-chromium-molybdenum alloys. The purpose of the present paper is to investigate the influence of alloyed molybdenum on the distribution of chloride ions in the passive film of an iron-chromium model alloy.

Experimental procedure

The composition of the two alloys used and of the two electrolytes are given in *Fig.2*. The temperature was 65 ^{o}C. The working electrode was a rotating disk, the rotation rate was 3000 rpm in all experiments. The alloys were mecanically polished and cathodically prepolarised before an experiment. The polarisation experiments were carried out in absence of air by applying a constant potential for three hours. The electrodes were then removed from the electrolyte, rinsed with distilled water and dried in a stream of argon. They were stored in a dessiccator for less than two hours before being transferred into the

UHV apparatus and analysed. A few experiments were performed in a glove box in presence of $^{18}O_2$ gas to study oxidation during sample cleaning and transfer.

metal	film formation electrolyte	analysis method	Cl^- in film	reference
Fe	phtalate	XPS	no	Khalil et al (2)
Fe	borate	AES	no	Janik-Czachor et al (3)
Fe	borate	SIMS	yes	Murphy et al (4)
Fe	borate+NaCl	SIMS	no	Goetz et al (5)
FeCr FeCrMo	NaCl	AES	no	Cieslak et al (6)
Fe26Cr	H_2SO_4 $H_2SO_4 + Cl^-$	AES	no yes	Mitrovic et al (7)
Ni	Na_2SO_4 +NaCl	AES,SIMS	yes	Mac Dougall et al (8)

Fig. 1 Investigations of Chloride Ions in Passive films by Surface Analysis Methods

Alloys: Fe-24Cr
Fe-24Cr11Mo

Electrolytes:

(A): 0.1 M H_2SO_4 + 0.4M Na_2SO_4
(B): 0.1 M H_2SO_4 + 0.4 M Na_2SO_4 + 0.12 M NaCl

Temperature: 65 C

Potentials: -650 mV nhe (CA)
+200 mV nhe (PL)
+500 mV nhe (PH)

Fig.2 Experimental Conditions

Surface analysis was performed in an PHI 590/550 AES/XPS system with CMA or in a PHI 545/Atomica Adida 3000-30 AES/SIMS system.

Details of the surface analysis instruments and procedures used are given elsewhere (9-11).

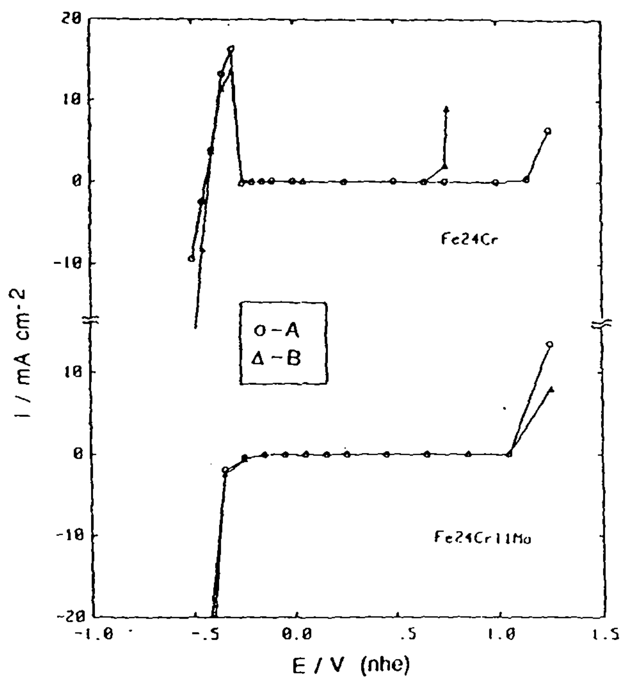

Fig.3 Polarisation Curves of Fe24Cr and Fe24Cr11Mo *in* Electrolytes A and B

Electrochemical Experiments

The polarisation curves of the two alloys in the electrolytes A and B are presented in *Fig.3*. The Fe24Cr alloy in electrolyte B exhibits pitting but not the Fe24Cr11Mo alloy. The latter is spontaneously passive in both electrolytes. The polarisation potentials for the surface analysis experiments were chosen at -650 mV (CA), +200 mV (PL) and +500 mV (PH) with respect to the standard hydrogen electrode. The first value corresponds to a cathodic polarisation (CA) and is used as a reference.The potential (PL) is in the low passive region at a potential below that for transpassive dissolution of pure molybdenum in the two electrolytes. The potential (PH) is in the high passive (PH) region above the potential of transpassive molybdenum dissolution. All applied potentials are negative to the value of the pitting potential of the Fe24Cr alloy in electrolyte B.

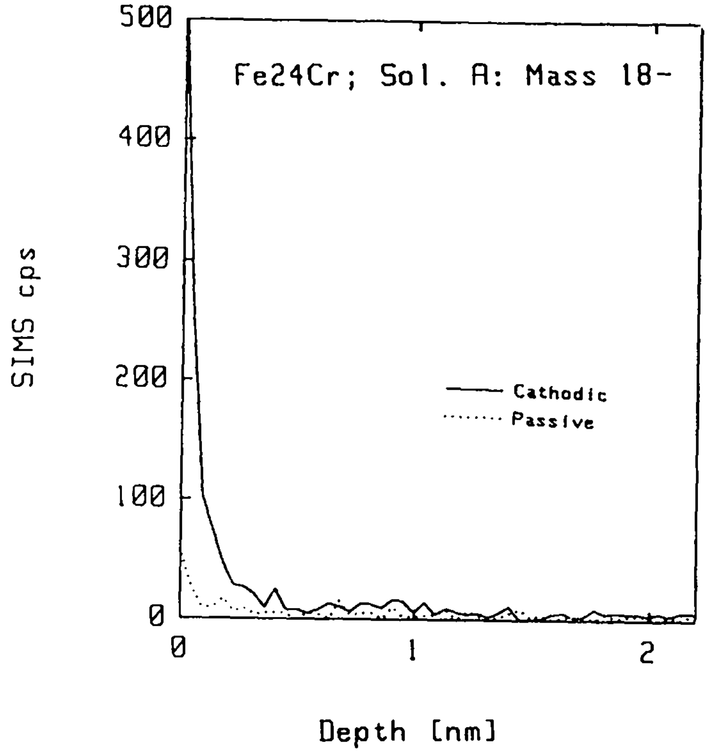

Fig.4 SIMS Depth Profile of ^{18}O on Fe24Cr Alloy Polarised at Potentials CA and PH in Electrolyte (A)

Experiments with $^{18}O_2$

AES, XPS and SIMS are ex-situ methods and, therefore, alteration of film composition during sample handling and transfer can not be excluded a priori (1). The occurrance or not of post-oxidation was investigated using the oxygen isotope ^{18}O. In these experiments polarisation was carried out in a glove box backfilled with nitrogen which contained approx 1250 ppm of oxygen 18 and 45 ppm of oxygen 16. In *Fig.4* the SIMS profiles of ^{16}O and ^{18}O for a cathodically polarised sample (CA) and an anodically polarised sample (PH) are shown. Both samples were exposed to the glovebox atmosphere after polarisation. The surface film formed on the cathodically polarised sample exhibits a much stronger ^{18}O signal than the film formed on the anodically polarised sample. *Fig.5* shows the intensity ratio $^{18}O/^{16}O+^{18}O$ for the same experiments. By plotting the intensity ratio rather than the count rate possible matrix effects are eliminated. The data confirm the difference in ^{18}O signal intensity between the two samples apparent in Fig.4. It is concluded that the film observed on the cathodically polarised sample is formed after emersion by reaction with oxygen or with rinsing water, while the film observed on the anodically polarised sample (PH) is formed in the electrolyte during anodisation and is not further oxidised after removal from the solution.

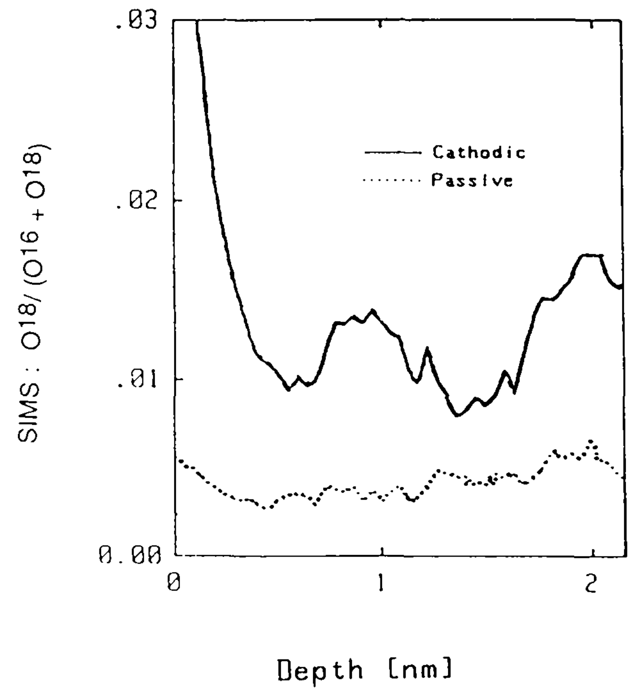

Fig. 5 SIMS Intensity Ratio $^{18}O/\ ^{16}O + ^{18}O$ for Fe24Cr Polarised in Electrolyte (A) at Potentials CA and PH

Film thickness

The film thickness was estimated from XPS data using deconvolution of the metallic and oxide part of the Cr 2p3/2 peak (10,11). The results obtained for the two alloys polarised in the electrolytes (A) and (B) are shown in *Fig.6*. The apparent film thickness is not significantly affected by the presence or not of chloride ion in the electrolyte or of molybdenum in the alloy. The film thickness of the samples that were cathodically polarised (CA) is that of the oxide film formed spontaneously during sample handling and transfer. The films formed during anodic passivation are only slightly thicker than the films observed on the cathodically polarised samples.

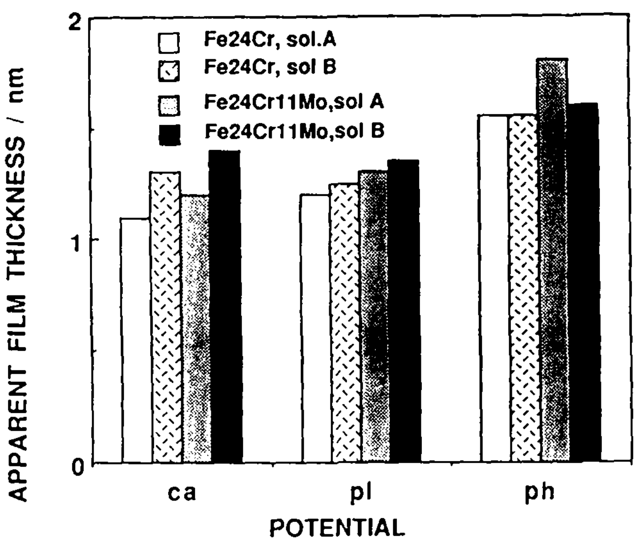

Fig.6 Apparent Film Thickness Calculated from Cr $2p_{3/2}$ XPS Peaks

Chloride ion analysis

AES profiles were determined using a rastered 1 KeV Kr^{+} ion beam. Measured profiles were numerically corrected for the effect of the finite escape depth of the Auger electrons (10,11). *Fig. 7* shows the chloride depth profile (LMM peak at 184 eV) measured in solution B on the Fe24Cr alloy. The cathodically polarised sample shows a measurable chloride signal only at the surface. This signal is attributed to adsorbed anions that remained after the rinsing procedure. The samples polarised in the passive potential region exhibit a chloride maximum inside the film, at an apparent depth of 0.3-0.4 nm. This suggests that the passive films formed under these conditions contained chloride. A rough estimation using sensitivity factors indicates a maximum concentration of less than 1 at%. On the Fe24Cr11Mo alloy an AES chloride profile could not be determined because of overlap of the LMM Cl peak at 181 eV with the MNV Mo peak at 186 eV.

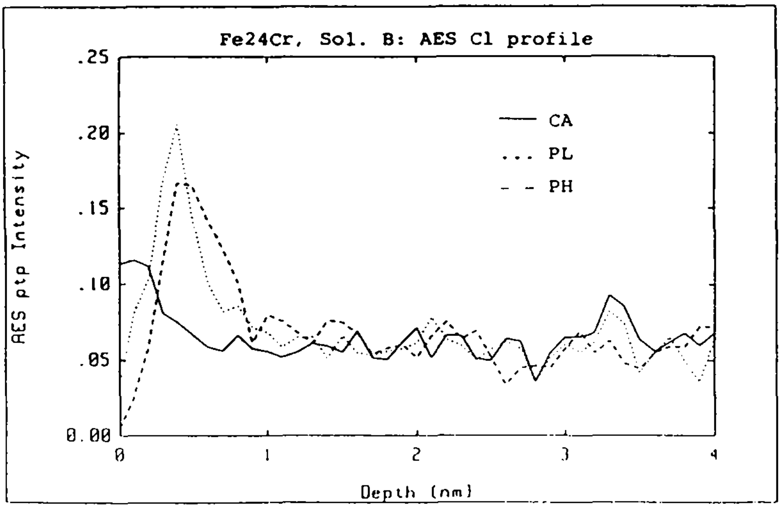

Fig. 7 AES Depth Profile of Cl Peak on Fe24Cr Alloy Polarised at CA, PL and PH

Further information was obtained from SIMS profiles. *Fig.8* shows negative SIMS profiles of the peak of the mass/charge ratio of 35 corresponding to chloride ion. The intensity of the profiles is normalised to the same height in order to better show the differences in depth distribution. The SIMS profiles of the Fe24Cr alloy qualitatively confirm the results of the AES depth profile analysis. On the cathodically polarised sample chloride is observed only at the oxide surface but on the passivated samples a chloride maximum occurs within the film.

Analogous data for the Fe24Cr11Mo alloy reveal no significant difference in the position of the Cl maximum between the cathodically polarised sample CA and the anodically polarised sample PL. On the other hand, the Cl maximum for the anodically polarised sample PH is shifted slightly to the right and is situated at a depth of approx 0.2-0.3 nm. This is less than the value of 0.3-0.4 nm found by SIMS and by AES for the Fe24Cr alloy. It appears from these data that less chloride ion is present in the passive film formed on the molybdenum containing alloy than on the alloy without molybdenum.

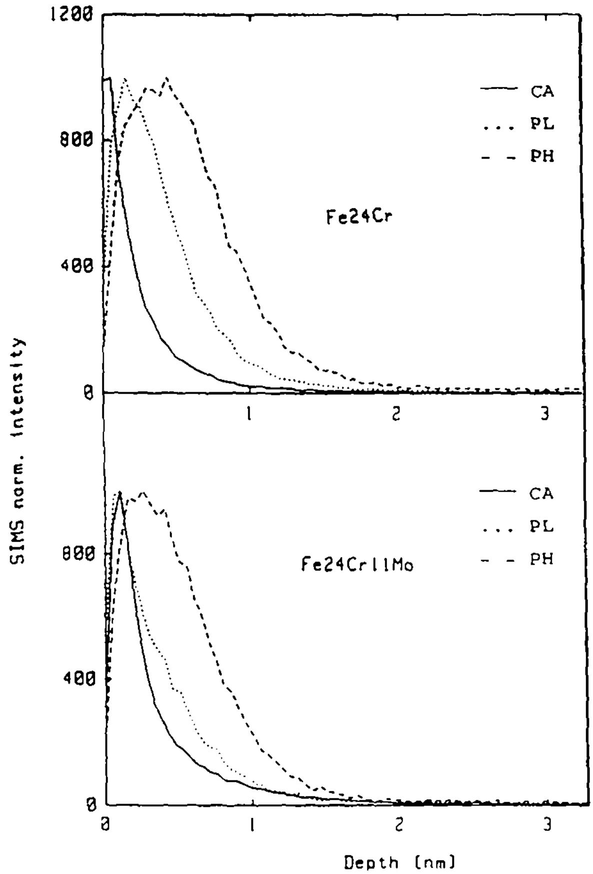

Fig.8 Normalised SIMS Cl^- Profiles measured on Fe24Cr and Fe24Cr11Mo polarised at CA, PL, and PH.

Discussion

The analysis of chloride ion in passive films on iron-chromium alloys is very delicate because surface analytical data are easily obscured by artifacts. First of all, in presence of molybdenum AES can not be used reliably for the study of chloride profiles because of the interference between the Cl 181 eV peak and the Mo 186 eV peak. On the other hand, XPS has a relatively low sensitivity for chloride. SIMS, in principle, is best suited but if used alone offers several difficulties of interpretation (9). Indeed, measured SIMS intensities are heavily dependent on matrix effects and are not necessarily proportional to the concentration of the element analysed. To avoid these difficulties different methods were employed in the present study and all data were interpreted based on relative measurments using the cathodically polarised sample as a reference. The obtained results, therefore, are believed to be quite reliable.

The SIMS and AES data show that chloride is present in the passive film on the Fe24Cr alloy formed in the chloride containing electrolyte B. The shape of the chloride concentration profiles suggests that the chloride ions are situated in the outer part of the passive film, but distinctly deeper than adsorbed ions. The film formed on the Fe24Cr11Mo alloy apparently contains less chloride than that on Fe24Cr. The observed difference can perhaps explain the well known improvement of the stability of passive films on iron-chromium alloys by molybdenum but a detailed discussion of various hypotheses is beyond the scope of the present paper.

The surface analysis methods used here give an average value over the analysed area rather than the local concentration on a microscopic level. It is, therefore, not possible from these data to decide whether the chloride ions are distributed uniformly on the analysed area or whether their presence is restricted to certain areas such as local defects or micropores. The shape of the chloride profiles showing almost no chloride ion in the inner part of the film and at the film-metal interface seems to exclude that film rupture-repassivation events exposing briefly small parts of the bare metal surface are at the origin of the observed presence of chloride ions in the films.

Conclusions

-Under the conditions of the present study the apparent thickness of the passive film measured by XPS on Fe24Cr is not significantly affected by the presence of chloride ion in the electrolyte or of molybdenum in the alloy.
-Passive films formed in the chloride containing electrolyte B on Fe24Cr contain more chloride than those formed under identical conditions on Fe24Cr11Mo.
-The depth distribution of the chloride ions in the film is non uniform, with a higher concentration in the outer part of the film.

Acknowledgement

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